

LA GRANDE RIVIERE, QC, Canada

WMO: 718270

Lat: 53.630N

Lon: 77.700W

Elev: 639

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.1	-26.5	-38.3	0.6	-29.8	-34.6	0.8	-26.3	25.0	0.4	22.4	0.2	6.9	200	0.801

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	80.2	63.1	76.4	61.0	72.8	59.6	66.2	75.2	64.0	71.5	61.9	69.1	12.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	88.9	68.9	61.1	82.3	66.9	58.9	76.1	65.3	31.4	74.8	29.7	71.4	28.1	69.1	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	20.6	18.5	DB	-34.4	86.8	4.1	4.0	-37.4	89.7	-39.8	92.0	-42.0	94.2	-45.0	97.1	(4)
(5)			WB	-34.6	69.9	4.1	2.6	-37.6	71.7	-39.9	73.2	-42.2	74.7	-45.1	76.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.8	-6.5	-5.2	7.7	23.7	40.4	53.1	58.6	56.7	48.1	37.2	23.1	6.6
	DBStd	25.06	13.05	12.54	14.26	11.72	11.03	9.81	7.35	7.06	7.57	7.36	8.60	12.65
	HDD50	8529	1752	1545	1311	791	343	79	12	15	126	404	806	1345
	HDD65	13272	2217	1965	1776	1240	765	373	224	274	510	862	1256	1810
	CDD50	793	0	0	0	0	45	172	278	223	68	7	0	0
	CDD65	61	0	0	0	0	2	15	25	17	2	0	0	0
	CDH74	604	0	0	0	0	45	186	235	126	12	0	0	0
	CDH80	118	0	0	0	0	8	39	52	19	0	0	0	0
Wind	WSAvg	9.3	8.9	8.8	9.1	9.3	9.4	9.4	9.0	9.1	10.1	9.7	9.8	9.2
Precipitation	PrecAvg	28.2	1.3	0.8	1.1	1.1	1.8	2.4	3.4	3.9	4.2	3.5	2.8	1.7
	PrecMax	35.2	3.4	1.7	1.9	2.3	3.6	5.0	5.7	6.4	9.2	6.2	5.5	2.9
	PrecMin	21.9	0.3	0.4	0.2	0.2	0.6	1.1	1.3	1.6	2.0	2.1	1.1	0.9
	PrecStd	3.5	0.7	0.4	0.5	0.6	0.8	1.1	1.1	1.3	1.4	0.9	0.8	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.3	32.7	44.8	57.5	80.5	85.0	85.2	82.9	75.7	63.1	47.1	34.8
		MCWB	31.7	31.3	38.4	44.4	59.3	66.5	65.6	65.5	63.2	54.8	44.5	33.6
	2%	DB	26.3	26.6	38.5	49.0	71.8	79.2	79.9	77.5	69.1	56.6	40.4	31.6
		MCWB	25.5	25.0	33.8	39.9	55.4	61.4	63.1	63.4	59.7	52.1	38.8	31.1
	5%	DB	19.8	21.9	34.0	44.2	65.3	74.8	75.8	73.3	64.6	52.3	36.8	27.9
		MCWB	19.1	20.6	31.1	36.9	52.4	59.3	60.9	61.5	57.7	48.9	35.6	27.1
	10%	DB	13.7	15.1	29.9	40.6	59.3	70.4	72.2	69.2	60.5	48.0	33.8	24.2
		MCWB	13.0	13.8	27.1	34.8	48.7	56.8	59.7	59.5	55.0	45.1	32.8	23.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.7	31.6	39.8	45.2	61.4	68.4	70.0	68.2	65.5	57.3	45.2	33.8
		MCDB	32.1	32.7	44.3	55.0	73.7	81.5	80.2	75.5	72.4	61.8	46.6	34.3
	2%	WB	25.4	25.2	34.5	40.9	56.8	65.2	65.7	65.7	62.0	53.3	39.0	31.0
		MCDB	26.2	26.4	37.9	47.7	68.8	74.1	74.1	74.0	66.8	56.1	40.1	31.5
	5%	WB	19.2	20.6	31.2	37.8	53.1	61.5	63.4	63.4	59.0	49.2	35.7	27.3
		MCDB	19.8	21.9	34.0	43.5	64.3	70.1	70.9	70.7	63.3	52.1	36.7	27.9
	10%	WB	12.9	13.9	26.9	35.0	49.3	58.5	61.6	61.1	55.3	45.4	32.9	23.5
		MCDB	13.6	15.0	29.8	39.8	58.2	68.9	69.3	67.3	60.0	48.0	33.8	24.2
Mean Daily Temperature Range	5% DB	MDBR	14.5	17.7	19.9	18.1	19.6	21.6	19.3	17.1	14.2	10.3	9.3	11.8
		MCDBR	17.2	19.2	19.3	20.3	27.5	25.9	24.0	22.1	20.2	17.3	10.5	12.3
		MCWBR	17.2	18.4	16.4	13.6	15.4	13.2	11.6	10.9	13.1	13.2	9.7	12.5
	5% WB	MCDBR	17.3	19.2	19.6	19.1	26.4	22.9	20.1	18.7	18.4	15.9	10.3	12.1
		MCWBR	17.4	18.5	17.3	13.4	15.9	14.5	12.1	11.2	13.9	13.2	9.8	12.3
Clear-Sky Solar Irradiance	taub	0.245	0.264	0.280	0.329	0.346	0.356	0.370	0.359	0.347	0.305	0.273	0.225	
	taud	2.169	2.192	2.210	2.195	2.396	2.428	2.424	2.457	2.472	2.540	2.289	2.208	
	Ebn at Noon	231	266	287	285	286	283	277	274	263	254	219	219	
	Edh at Noon	20	28	34	40	35	34	34	31	27	20	18	16	
All-Sky Solar Radiation	RadAvg	293	646	1153	1606	1741	1864	1639	1289	865	478	215	180	
	RadStd	30	50	101	118	134	165	129	102	89	40	29	25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-23
(47) Regional (1 neighbor)	N/A	N/A	+2.09	-1.30	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page